Slurm Guide

CHANGELOG: v250215 - initial release
Instruction Staff: Find at 4248 >> Projects

This is a guide on how to use Slurm in the SoC Compute Cluster made by the CS4248 Teaching Team.

The SoC Compute Cluster may be used when you require extra compute, e.g. training your LLM for the group project. This guide aims to give you a brief overview of how to get started with the cluster - it is not an exhaustive resource for using the cluster.

References:

- https://dochub.comp.nus.edu.sg/cf/guides/compute-cluster/slurm-quick
- https://dochub.comp.nus.edu.sg/cf/guides/compute-cluster/slurm-info
- https://dochub.comp.nus.edu.sg/cf/guides/compute-cluster/gpu

You may view the GPU hardware configurations available on the SoC Compute Cluster here:
https://dochub.comp.nus.edu.sg/cf/guides/compute-cluster/hardware

Step 1: Prerequisites
You must either be in the SoC network or you can use the "SoC VPN" via FortiClient

Step 2: Logging in SoC Compute Cluster
Run:

```
ssh <username>@<hostname>
```

Replace `<hostname>` with `<xlogin.comp.nus.edu.sg>` (you can also specify if you want to login a specific node) and replace `<username>` with your SoC Unix username.

Note: these are both case sensitive

Step 3: Submitting a Simple Batch Job
Now that you have entered the SoC Compute Cluster, we will now show you how to submit Slurm jobs. Slurm is an open-source, fault-tolerant, and highly scalable cluster management and job scheduling system for Linux clusters, including our very own SoC Compute Cluster.

Slurm batch jobs are Unix shell scripts. Let's say we have an `.sh` file named `somejob.sh`. Then the content of the file would contain:

```
#!/bin/sh
```

```
srun <someprogram>
```

Replace `<someprogram>` with a program of your choice, e.g. for example you might want to run `python train.py` to run a train.py file that contains the script to train your model.

To submit the batch script to Slurm, run the following in the SoC Compute Cluster terminal:

```
sbatch somejob.sh
```

While your batch job is running, Slurm writes its terminal output to the `slurm-<jobid>.out` file by default. You can also specify your own output file using `-o <file-name>` (for stdout) and `-e <file-name>` (for stderr). For example, `sbatch -o myoutput.txt -e myoutput.err somejob.sh`.

Step 4: Adding Arguments

In Step 3, you have learned how to submit a simple batch job with default settings. In this case, you would be issued the default time limit for running a batch job, and be assigned a random CPU/GPU.

For many use cases, such as training a large model, these settings would not be enough. And we would have to configure the arguments of our batch script.

We have provided a simple bash script. Here, we have several arguments that may be useful for your reference:

- `job-name`: gives a name for the job to be executed, in this case we name it gpujob
- `gpus`: specifies the number of GPUs to be used, in this case we request for 1 GPU node (in this case the GPU allocation is random)
    - You can also specify a specific GPU to be used and format it as `<gpu-name>:<num-of-gpu>` GPU name can be found [here] under the "Slurm GPU name" column. For example, if I want to use 1 GPU of type "NVIDIA A100 with 80GB", then my #SBATCH argument would be `#SBATCH --gpus=a100-80:1`
- `time`: specifies the amount of time you would like to request to use the GPU (format in `HH:MM:SS`), in this case we request to use the GPU for 10 hours. The figure below outlines the max and default time you may use the GPUs for.

  Partitions

  There are two partitions (aka queues) to cater to different kind of job runtimes.

Partition Names	Max / Default Time	Priority Value	"Nice" Bonus
normal, gpu	3 hr / 15 mins	800	+200 for jobs 15 mins or less
long, gpu-long	5 days / 5 hours	200	+400 for jobs 5 hrs or less; +200 for jobs 1 day or less

- `mail-type`: specifies the actions that trigger an email message, in this case we set `BEGIN`, `END`, and `FAIL`, which means an email will be sent to you when the program starts and stops running, and if the program fails. This makes sure you can receive all updates via email :)

- `mail-user`: specifies the email address to send the notifications to

```
#!/bin/sh

#SBATCH --job-name=gpujob
#SBATCH --gpus=1
#SBATCH --time=10:00:00
#SBATCH --mail-type=BEGIN,END,FAIL
#SBATCH --mail-user=<your-email>

srun python some-gpu-program.py
```

Step 5: Other useful commands

There are also several commands that may be useful for you while using the SoC Compute Cluster:

- `sinfo`: Show the status of nodes and partitions in the Slurm cluster
- `squeue`: Show the Slurm job queue
    - Tip: you can use `squeue -u <username>` to specifically check for the status of your jobs in the queue
- `scancel <jobid>`: Cancel a specified job id (jobid can be found when job is queued, or from the `squeue` command). If the job is running, it will be aborted.
- `sacct`: List job accounting information. You can find jobs that are already over, including its status and exit code. This command can provide a lot of job and accounting information; please consult the man pages for more details.
- `sprio`: List factors that comprises jobs' scheduling priority.